

Anthony Bardou

RESEARCH SCIENTIST – MACHINE LEARNING AND OPTIMIZATION

+33 669 302 564 | anthony.bardou@epfl.ch | <https://abardou.github.io/> | [abardou](#) | [anthony-bardou-437b96170](#)

Research scientist specializing in machine learning, stochastic optimization, and adaptive decision-making under uncertainty. Experienced in designing scalable learning and optimization algorithms for noisy, dynamic, and decentralized systems, with strong emphasis on mathematical rigor, empirical evaluation, and real-world applicability. Publications at NeurIPS, ICLR, and IEEE TWC, with expertise spanning Bayesian optimization, online learning, probabilistic modeling, and efficient optimization of complex systems. Strong collaborator and mentor with experience leading research directions, supervising students, and contributing to open-source research tooling. Open to remote or hybrid roles; willing to relocate for the right position. French citizenship, allowed to work in the EU and Switzerland.

Skills

Research & Innovation	Algorithm Design, Theoretical Analysis, Experimental Evaluation, Technical Writing
Optimization & Learning	Online Learning, Bayesian Optimization, Reinforcement Learning, Stochastic Optimization
Engineering & Tools	PyTorch, NumPy/SciPy, Python, C/C++, Git, Linux
Applied Maths	Probabilistic Modeling, Bayesian Inference, Statistics, Linear Algebra, Calculus
Languages	French (native), English (full professional proficiency), Spanish (notions), Romanian (notions)

Experience

École Polytechnique Fédérale de Lausanne	Lausanne, CH
POSTDOCTORAL RESEARCHER - MACHINE LEARNING AND OPTIMIZATION	Oct. 2023 - Now
- Led research on scalable Bayesian optimization and adaptive learning algorithms for noisy, dynamic, and high-dimensional systems. - Developed theoretically grounded optimization methods with strong empirical performance on complex black-box learning and decision systems. - Designed and implemented research software in PyTorch and BoTorch for efficient experimentation and evaluation. - Collaborated with interdisciplinary researchers across machine learning, optimization, and systems domains. - Supervised and mentored 9 undergraduate and graduate students on independent research projects.	
Université Claude Bernard Lyon I	Lyon, FR
RESEARCH INTERN - NATURAL LANGUAGE PROCESSING	Dec. 2018 - Jun. 2019
- Researched attention and representation learning techniques for citation-aware scientific document understanding. - Improved classification F1 by 23 points on internal benchmarks.	

Education

École Normale Supérieure de Lyon	Lyon, FR
PHD IN MACHINE LEARNING AND WIRELESS NETWORKS	Oct. 2020 - Sep. 2023
- Designed zeroth-order optimization algorithms for high-dimensional noisy objectives in wireless networks. - Achieved significant improvements of most KPIs (throughput, fairness, latency) in wireless networks (Wi-Fi, cellular).	

Honors & Awards (selection)

2025-2026	NeurIPS Top Reviewer, recognized for high-quality, rigorous evaluation of research contributions (originality, technical correctness, significance)	Vancouver, CA & San Diego, US
2025	Best Ph.D. Thesis, for the quality, originality, and rigor of the research contribution	France
2022	Best Paper Award, 25th International Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems	Montreal, CA

Open-Source Projects

An Off-the-Shelf Kernel for Encoding Invariance in Bayesian Optimization	Feb. 2026	abardou/max-kernel
W-DBO: Wasserstein Distance-Based Dynamic Bayesian Optimization	May 2024	WDBO-ALGORITHM/wdbo_algo
DuMBO: High-Dimensional, No-Regret Bayesian Optimization	Feb. 2023	abardou/dumbo

Talks (selection)

April 2026	Invited Speaker , LIRA Lab (UFRJ)	Rio de Janeiro, BR
May 2025	Invited Speaker , Secondmind Research Seminar	Cambridge, UK
Jun. 2024	Invited Speaker , INRIA Maracas (INSA Lyon)	Lyon, FR

Publications (selection)

ANTHONY BARDOU, ANTOINE GONON, ARYAN AHADINIA, PATRICK THIRAN. Symmetry-Aware Bayesian Optimization via Max Kernels. *Proc. ICLR'26*. *Rio de Janeiro, Brazil*. Apr. 2026

ANTHONY BARDOU, JEAN-MARIE GORCE, THOMAS BEGIN. Assessing the Performance of NOMA in a Multi-Cell Context: A General Evaluation Framework. *IEEE Transactions on Wireless Communications*. Jul. 2025.

ANTHONY BARDOU, PATRICK THIRAN, GIOVANNI RANIERI. This Too Shall Pass: Removing Stale Observations in Dynamic Bayesian Optimization. *Proc. NeurIPS'24*. *Vancouver, Canada*. Dec. 2024

ANTHONY BARDOU, PATRICK THIRAN, THOMAS BEGIN. Relaxing the Additivity Constraints in Decentralized No-Regret High-Dimensional Bayesian Optimization. *Proc. ICLR'24*. *Vienna, Austria*. May 2024.

ANTHONY BARDOU, THOMAS BEGIN. INSPIRE: Distributed Bayesian Optimization for ImproviNg SPatial REuse in Dense WLANs. *Proc. MSWiM'22*. *Montréal, Canada (QC)*. Oct. 2022. 10.1145/3551659.3559050. **BEST PAPER AWARD**.

External Accolades

Prix de thèse du GDR RSD : des travaux sur l'optimisation récompensés	Jan. 2025	INS2I (CNRS)
Lauréat du prix de thèse GDR RSD - Édition 2024	Sep. 2024	GDR RSD (CNRS)
L'optimisation bayésienne au service des performances du WiFi	Apr. 2023	INS2I (CNRS)

Academic Service

REVIEWS

2025	International Conference on Learning Representations , International Conference
2025	Scientific Reports , Peer-Reviewed Journal (Nature)
2025	Transactions on Machine Learning Research , Peer-Reviewed Journal
2025	International Conference on Machine Learning , International Conference
2024	Computer Networks , Peer-Reviewed Journal (Elsevier)
2023	Neural Information Processing Systems , International Conference
2023	Computer Communication , Peer-Reviewed Journal (Elsevier)
2023	International Symposium on Information Theory , International Conference (IEEE)

TEACHINGS

Fall 2024-25	Stochastic Modeling , 60 hours @ EPFL (Bachelor)	Lausanne, CH
Spring 2023	Preparation to the agrégation of Computer Science , 32 hours @ ENS Lyon (Master)	Lyon, FR
Spring 2022	Numerical Methods , 15 hours @ UCBL (Bachelor)	Lyon, FR
Spring 2022	Optimization , 15 hours @ UCBL (Bachelor)	Lyon, FR
Fall 2021	Introduction to Artificial Intelligence , 12 hours @ UCBL (Master)	Lyon, FR
Fall 2021	Optimization and Operations Research , 18 hours @ UCBL (Master)	Lyon, FR

SUPERVISION AND MENTORING

Spring 2026	Marin Philippe , Master Thesis @ EPFL	Lausanne, CH
Spring 2025	Robin Jaccard , Master Thesis @ EPFL	Lausanne, CH
Spring 2025	Adrien Deschenaux , Bachelor Semester Project @ EPFL	Lausanne, CH
Spring 2025	Aryan Ahadinia , Master Semester Project @ EPFL	Lausanne, CH
Fall 2024	Antoine Merel , Master Semester Project @ EPFL	Lausanne, CH
Fall 2024	Nahuel Plomb Veuve , Bachelor Semester Project @ EPFL	Lausanne, CH
Spring 2024	Emanuele Mengoli , Master Thesis @ EPFL (from École Polytechnique)	Lausanne, CH
Spring 2024	Giovanni Ranieri , Bachelor Semester Project @ EPFL	Lausanne, CH
Spring 2023	Hugo Martel , Bachelor Research Internship @ ENS Lyon	Lyon, FR